

on time of RL1.

The circuit works off 12V DC power supply. Assemble it on a general-purpose PCB. After construction, mount the laser diode and the photo-

transistor on opposite sides of the doorframe and align them such that the light beam from the laser diode falls on the phototransistor directly. The motor connected to the pole of

relay contacts is the one used in electromechanical door-opener assembly. If you want to use a DC motor, replace mains AC connection with a DC power supply. ●

IR RECEIVER MODULE TESTER

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Here is a tester for on-board testing of IR receiver modules used for remote control of TV sets and VCD players. The circuit is very simple and can also function as a remote tester.

IR receiver modules are miniature IR receivers sensitive to pulsed infrared rays. These have a pin photo-diode and a preamplifier stage enclosed in an epoxy case that acts as the IR filter. Internally, the module has an AGC, band-pass filter, demodulator and control circuit. Its output has a bipolar transistor with 80- to 100-kilo-ohm resistor in the collector. Normally, the collector output of the transistor is high and gives 5V at 5

mA. The output of the module is active-low and hence it sinks current when the pin photo-diode senses the presence of pulsed IR rays.

The IR receiver module is designed with high immunity against ambient light and is capable of continuous data transmission at up to 2400 bps or higher. The band-pass filter and AGC suppress unexpected noise to avoid false triggering. The module responds to the IR beam only if its carrier frequency is close to the centre frequency of the band pass.

Working of the circuit is simple. Three mini crocodile clips are used to connect the circuit to the positive, negative and output of the module. If the module is properly working, its output



Fig. 2: Pin configuration of TSOP1738

remains 5 volts. This makes the cathode of LED1 high. So LED1 doesn't glow and the buzzer remains silent.

When you focus the remote handset onto the IR receiver and press any switch, the output of the IR receiver sinks current. So LED1 starts flashing and the buzzer beeps in sync with the pulsations of the IR beam. On the other hand, if your IR receiver module is faulty, the output of the module does not sink current when you focus the remote handset towards the module and press any switch. So neither LED1 flashes, nor the buzzer beeps in sync with the pulsations of the IR beam.

Power to the circuit is obtained from a 9V PP3 battery and regulated to 5 volts by zener diode ZD1. Most of the IR receiver modules work only between 3 and 6 volts. Storage capacitor C1 releases current to make LED1 flash brightly. (EFY Note. We had used a TSOP1738 IR receiver module while testing. Fig. 2 shows the pin configuration of TSOP1738.)

Assemble the circuit on a small piece of matrix board and enclose in a small cabinet. Use a high-brightness red LED and a small buzzer for audio-visual indication. Connect points A, B and C to the crocodile clips using red, black and blue wires to connect to the pins of the module easily. For easy identification of pins, the pin assignment (front view) of some common IR receiver modules is shown in the table. ●

Pin Assignment of Some Common IR Receiver Modules

IR receiver module	Pin 1	Pin 2	Pin 3
TSOP 1730, 33, 36, 37, 38, TSOP 3856, TK 19-907, TK 19931, TSOP 1236, TSOP 1138, TFMM 5380	Ground	Vcc	O/P
SFH 505, 505A	Vcc	Ground	O/P
TK1738, SFH506, SFH5110, ISIV60, IS1460, TSOP1838, RPM7238, RPM7100	O/P	Ground	Vcc
GP1U561X	Vcc	O/P	Ground

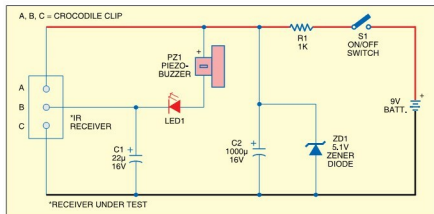


Fig. 1: IR receiver module tester circuit